Reg. No: | | | | | | | | | | | |

SRI RAMAKRISHNA ENGINEERING COLLEGE, COIMBATORE

(Autonomous Institution, Reaccredited by NAAC with 'A+' Grade Approved by AICTE and Permanently affiliated to Anna University, Chennai)

AUTONOMOUS EXAMINATIONS - JAN'2025

COMMON TO B.E – CSE / B.TECH – IT & M.TECH - CSE

20MA203 - LINEAR ALGEBRA AND DIFFERENTIAL CALCULUS

Answer ALL questions

Duration: 3 Hours **Maximum: 100 Marks**

PART - A **(7x 1 = 7 Marks)**

1. **CO1 The Eigen values of a real symmetric matrix are _______.**

   a) Real b) Imaginary c) Singular d) Non Singular

2. **CO2 Let V be a vector space and let x, y be vectors in V, then x + y = y implies_____.**

   a) x = 1 b) x = 0 c) ax = 1 d) x = y

3. **CO2 If A is an $m \times n$ matrix, then which of the following is correct?**

   a) dim R(A)≠dim C(A) b) dim R(A)>dim C(A)

   c) dim R(A)=dim C(A) d) dim R(A)<dim C(A)

4. **CO3 Two vector spaces over the same field are isomorphic if they have same ______.**

   a) dimension b) sub space c) basis d) linearity

5. **CO4 What is the radius of curvature of the curve $x + y = 1$ ____.**

   a) 3 b) 0 c) 2 d) ∞

6. **CO3 A function f : $S \rightarrow T$ is invertible if and only if it is _______.**

   a) Injective b) Bijective c) Surjective d) Both (a) and (c)

7. **CO5 If $u = f(x, y)$ is an implicit function in x and y then $\frac{dy}{dx} =$ ______.**

   a) $\frac{f_x}{f_y}$ b) $\frac{f_y}{f_x}$ c) $-\frac{f_x}{f_y}$ d) $-\frac{f_y}{f_x}$

Fill in the blanks: **(3x 1 = 3 Marks)**

8. **CO1** The characteristic equation of the matrix $\begin{pmatrix} 3 & 1 \\ 2 & 3 \end{pmatrix}$ is _______.

9. **CO3** A function $T: V \to W$ is called ________ transformation if T(x) = x for all $x \in V$.

10. **CO5** If u and v are functions of x and y then $\frac{\partial(u,v)}{\partial(x,y)} \times \frac{\partial(x,y)}{\partial(u,v)} =$ ________.

PART - B **(5x 3 = 15 Marks)**

11. **CO1** Find the Eigen values of A^{-1} if the two of the Eigen values of $A = \begin{pmatrix} 2 & 2 & 1 \\ 1 & 3 & 1 \\ 1 & 2 & 2 \end{pmatrix}$ are equal to one each.

12. **CO2** Show that the vectors (1,1,1), (0,1,1) and (0,0,1) span R^3.

13. **CO3** Show that g is linear if $g: R \to R$ defined by $g(x) = x$.

14. **CO4** Find the radius of curvature for the curve $y = e^x$ at (0,1).

15. **CO5** Find the Jacobian $\frac{\partial(u,v)}{\partial(x,y)}$ for the following function $u = \frac{y^2}{x}$ and $v = \frac{x^2}{y}$.

PART - C **(5 x 15 = 75 Marks)**

Q.No: 16 Compulsory Question

Q.No: 17 to 22 Answer any <u>FOUR</u> Questions

<u>Compulsory Question:</u>

16. **CO1** Solve and reduce the quadratic form $2x_1^2 + 2x_2^2 + 2x_3^2 - 2x_1x_3$ to canonical form by orthogonal reduction. Also find rank, index, signature and nature. (15)

17. **CO2** Determine the bases for the row space, column space, null space, rank and nullity of the matrix (15)

$$A = \begin{pmatrix} 2 & 4 & -3 & -6 \\ 7 & 14 & -6 & -3 \\ -2 & -4 & 1 & -2 \\ 2 & 4 & -2 & -2 \end{pmatrix}$$

18. **CO3** i) Let A be a matrix on R^2 and basis S of R^2 be (8)

$A = \begin{bmatrix} 3 & -2 \\ 4 & -5 \end{bmatrix}$ and $S = (u_1, u_2) = \left\{ \begin{bmatrix} 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 2 \\ 5 \end{bmatrix} \right\}$. Find matrix representation with respect to

- basis S
- standard basis

CO3 ii) Find the matrix representation of $[T]_\alpha$ *and* $[T]_\beta$ of each of the following linear transformation on R^3 with respect to the standard basis $\alpha = \{e_1, e_2, e_3\}$ and another $\beta = \{e_3, e_2, e_1\}$, $T(x, y, z) = (2y + z, x - 4y, 3x)$. And also find $[T]_\alpha^{\ \beta}$ of the linear transformation T. (7)

19. **CO4** i) Construct the equation of the evolute of the parabola $y^2 = 4ax$. (8)

CO4 ii) Find the radius of curvature for the curve $\sqrt{x} + \sqrt{y} = 1$ at $\left(\frac{1}{4}, \frac{1}{4}\right)$. (7)

20. **CO5** i) A thin closed rectangular box is to have one edge equal to twice the other, and a constant volume $72\,m^3$. Find the least surface area of the box. (8)

CO5 ii) Expand $e^x \cos y$ in powers of x and y as for as terms of the third degree about (0,0). (7)

21. **CO1** i) Find the Eigen values and eigen vectors of the matrix $\begin{bmatrix} 4 & 2 & -2 \\ -5 & 3 & 2 \\ -2 & 4 & 1 \end{bmatrix}$. (8)

CO3 ii) Show that the vectors (1,2,1) (2,9,0) and (3,3,4) in R^3 form a basis. (7)

22. **CO4** i) Find the equation of the circle of curvature for the curve $xy = c^2$ at (c, c) (8)

CO5 ii) If $u = \log(x^2 + y^2 + z^2)$, prove that $\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} + \frac{\partial^2 u}{\partial z^2} = \frac{2}{x^2 + y^2 + z^2}$. (7)

*****